

Module-1. (Fundamentals)

↳ Not for Gate but for other subjects.

- Sequence & Series
- Modular Arithmetic
- Logarithmic
- Importance of log in CS
- Finding roots of polynomials
- Proof Techniques.

Lecture
I

⇒

Summation:

$$\sum_{x=1}^n f(x) \xrightarrow{\text{Unrolling}} f(1) + f(2) + \dots + f(n)$$

↳ Unrolled form

Not $f(1+2+3+\dots+n) \times$

Like loop. ↙

Here x is dummy variable, it can be i, j , anything

$$\sum_{i=1}^n f(i) = \sum_{j=1}^n f(j) \neq \sum_{j=1}^n f(x)$$

→ Shifting indices of summation.

$$\text{Ex: } \sum_{j=0}^3 2j = \sum_{j=1}^a b, \text{ then } a=? , b=?$$

$$\text{as } i-j=1$$

$$\text{So, } a-3=1$$

$$\underline{a=4}$$

$$b = j = i-1$$

So

$$b = (i-1)$$

$$\text{So } \sum_{j=0}^3 2j = \sum_{i=1}^4 2(i-1)$$

Ex: 2

$$\sum_{i=2}^{10} i(i-2) = \sum_{j=1}^{10} (j-1)(j+1) = \sum_{j=1}^{10} (j^2-1)$$

as $j = i-1$
 replace i with $j+1$

Ex: 3

$$\sum_{i=1}^n \frac{i-1}{m}$$

start summation from 0.

⇒

$$\sum_{j=0}^{n+1} \frac{j+1-1}{m}$$

$$= \sum_{j=0}^{n+1} \frac{j}{m}$$

GO (j = i-1)

CLASSES

start summation from 2

$$\Rightarrow \sum_{j=2}^{n+1} \frac{j-1-1}{m}$$

(j = i+1)

$$= \sum_{j=2}^{n+1} \frac{j-2}{m}$$

$$\Rightarrow \sum_{j=i}^n j = \sum_{k=0}^{n-i} i+k$$

Ques: Use intuition to express following in notation of summation:

a). $1 + 20 + 400 + 8000$

$$\rightarrow \sum_{i=0}^3 20^i$$

b). $-3 - 1 + 1 + 3 + 5 + 7$

$$= \sum_{i=-3}^7 2i+1$$

★ Summation Identities:

$$\rightarrow \sum_i a x_i = a \sum_i x_i$$

$$\rightarrow \sum_i (x_i + y_i) = \sum_i x_i + \sum_i y_i$$

→ Rearrangement

$$\rightarrow \sum_{i=a}^b f(x_i) = \sum_{i=0}^b f(x_i) - \sum_{i=0}^{a-1} f(x_i)$$

$$\sum_{k \in A} k = a|A|, \quad a \text{ is constant.}$$

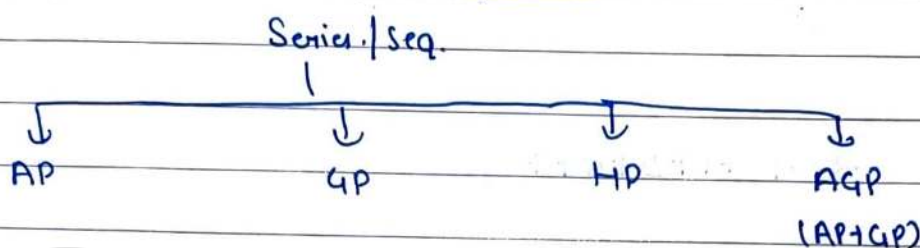
$$\rightarrow \sum_{i=a}^b k \Rightarrow k \sum_{i=a}^b 1 \Rightarrow k(i+1) - k(a) = k(b-a+1)$$

→ $b-a+1$ times

, k is constant.

Lecture
IISequences & Series:

If $a_1, a_2, a_3, \dots, a_n$ is a sequence of n terms then the corresponding series is $a_1 + a_2 + a_3 + \dots + a_n$. Series is sum of n terms of sequence.



→ Arithmetic Progression: Seq. of terms which follows a specific common difference.

Ex: $0, 2, 4, 6, 8, \dots$ is A.P. as it is following some common diff. i.e. 2.

In AP, each term differs from the next term by the same, fixed quantity, called common diff. (d)

Ex:

Series

First term (a)

Common Diff (d)

i) $7, 11, 15, 19, 23, \dots$

7

$11 - 7 = 4$

ii) $\frac{1}{6}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \dots$

$\frac{1}{6}$

$\frac{1}{3} - \frac{1}{6} = \frac{1}{6}$

iii) $a+2b, a+b, a, a-b, a-2b, \dots$

$a+2b$

$-b$

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Ex: If a_n represents n^{th} term of series, and common diff in series is d and $a_1 = a$ = first term, then,

w.k.T. $a_2 = a + d$,

& $a_3 = a_2 + d = a + 2d$

then, $a_n = a + (n-1)d$ $\hookrightarrow n^{\text{th}}$ term of AP.

Also $a_n = a_{n-1} + d$

Ex: If first 3 terms of AP are 8, 5, 2, then what is the 16th term?

$\rightarrow$ Here $a = 8$, $d = a_2 - a_1$

$= 5 - 8 = -3$

$a_{16} = 8 + (16-1)(-3)$

$= 8 + (15)(-3)$

$= -45 + 8 = -37$

Ans

Ex: ~~Given~~

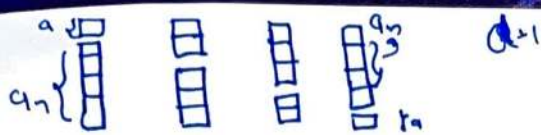
Ex: Given that $2x+5$ and $6-x$ are first two terms of A.P., then what is x .

$\Rightarrow 2x + d = 5$ --- (i)

$6 - x - d = 5$ --- (ii)

Solving (i) & (ii)

$x = 4$, $d = -3$



$$\begin{matrix} 1 \leftrightarrow a_n \\ a \leftrightarrow 1 \end{matrix} \} \text{Conventions}$$

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→ Sum of first n terms in AP:

$$a_1 + a_2 + a_3 + \dots + a_n$$

It looks like this in number line



$$a_1 + a_2 + a_3 + \dots + a_{n-2} + a_{n-1} + a_n$$

$$\Rightarrow \text{So } a_1 + a_n = a + a_n \quad (\text{First \& last})$$

$$a_2 + a_{n-1} = a + d + a_n - d = a + a_n \quad (\text{Second \& Second last})$$

$$a_3 + a_{n-2} = a + 2d + a_n - 2d = a + a_n$$

→ Obviously can also check using AP

$$\text{Ex: } 1, 3, 5, 7, 9, 11$$

$$1 + 11 = 12 = 9 + 3 = 12 = 5 + 7 = 12$$

So $\frac{n}{2}$ such pairs will be there (if n is even)

Then sum will be

$$\frac{n}{2} \times (a + a_n)$$

$$\text{Now } \frac{n}{2} \times (a + a_n + (n-1)d) \quad (\text{As } a_n = a + (n-1)d)$$

$$= \frac{n}{2} (a + a + nd - d) \Rightarrow \frac{n}{2} (2a + (n-1)d)$$

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→ If a, b, c, d, e is AP with c.d. x , then a, c, e is also A.P. with c.d. $2x$.

Also $\Rightarrow c = \frac{a+e}{2}$ as

$$c - a = e - c \Rightarrow 2c = a + e \Rightarrow c = \frac{a+e}{2}$$

Now in sum of AP case if n is odd then we will have $\frac{n-1}{2}$ pairs and we will be left without middle term which will be $\frac{a+t_n}{2}$. (Mid of series)

So sum will be:

Represents Sum of n no. in series $\leftarrow S_n = \text{Sum} = \frac{(n-1)}{2} (a+t_n) + \frac{a+t_n}{2}$ (if n is odd)

$$= \frac{a+t_n}{2} (n-1+1)$$

$$= \frac{n(a+t_n)}{2} = \text{Same as Even}$$

So sum always is $\frac{n(a+t_n)}{2}$ if n is odd/even

Ex: In A.P. 1, 3, 5, ...

Find sum of 15 terms

$\rightarrow d = 3 - 1 = 2$

$a = 1$

$$S_{15} = \frac{15}{2} (1 + a_n) \Rightarrow \frac{15}{2} (1 + a + (n-1)d) = \frac{15}{2} (1 + 1 + 14 \cdot 2)$$

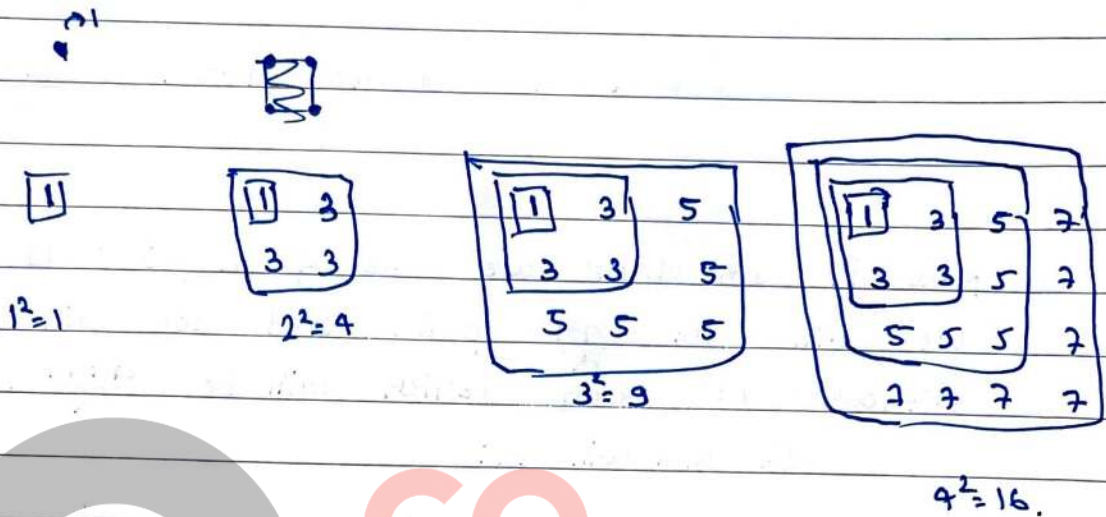
$$\Rightarrow \frac{15 \cdot 30}{2} = 225$$

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→ Sum of starting ^{consecutive} n odd numbers is n^2 (That's why here it was $15^2 = 225$)

Visual Repr. (Not imp) ↓



and so on.

★ Imp formulas:

$$1 + 2 + 3 + 4 + 5 + \dots + n = \frac{n(n+1)}{2}$$

↳ Proof

$$\frac{n}{2}(n+1) \rightarrow \frac{n(n+1)}{2}$$

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Remember

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

Lecture - III

⇒ Geometric Progressions:

10, 100, 1000, ... is an ex. of G.P.

1 $\xrightarrow{\times 2}$ 2 $\xrightarrow{\times 2}$ 4 $\xrightarrow{\times 2}$ 8 $\xrightarrow{\times 2}$ 16This is called common ratio (was diff. in A.P.) $\rightarrow (r)$ $= \frac{a_2}{a_1} = \frac{a_3}{a_2}$

Generally G.P. is:

 $a, ar, ar^2, ar^3, \dots$ Now $a_1 = a,$ $a_2 = ar$ $a_3 = ar^2$

So

$$a_n = ar^{(n-1)}$$

 $\rightarrow n^{\text{th}}$ term of G.P.

$$r = \frac{a_{k+1}}{a_k}$$

Ex: Given first two terms in G.P. are 5 & 12, what is the 7th term

$$\Rightarrow a = 5, r = \frac{12}{5}$$

$$a_7 = 5 \times \left(\frac{12}{5}\right)^{7-1} \Rightarrow 5 \times \left(\frac{12}{5}\right)^6$$

Ques: How many terms are there in G.P.

2, 4, 8, ..., 128?

$$\Rightarrow a = 2, r = 2, n = ?, a_n = 128$$

$$128 = 2 \times (2)^{n-1}$$

$$128 = 2^n$$

$$n = 7$$

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⇒ Sum of GP: Sum of first n terms of GP.
 S_n can be given as:

$$S_n = a + ar + ar^2 + \dots + ar^{n-1} \quad \text{--- (i)}$$

$$rS_n = ar + ar^2 + ar^3 + \dots + ar^n \quad \text{--- (ii) (multiply } r \text{ both side)}$$

Now

$$S_n - rS_n = a + ar^n \quad \text{--- (i) - (ii)}$$

$$S_n(1-r) = a(1-r^n)$$

$$S_n = \frac{a(1-r^n)}{(1-r)}$$

So

$$S_n = \frac{a(1-r^n)}{(1-r)}$$

→ for G.P.

i.e. $r \neq 1$

↓

if $r=1$ then $S_n = na$
obviousEx:

Sum of GP

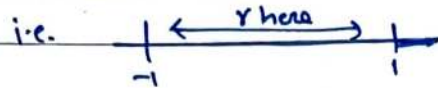
$$1 + 3 + 3^2 + \dots + 3^{10}$$

$$\Rightarrow a=1, r=\frac{3}{1}=3, n=11$$

$$S_{11} = \frac{1(1-3^{11})}{1-3} \Rightarrow \frac{3^{11}-1}{2}$$

if $n=54$ then $n=2$ strictly on
 but if $n^2=4$ then $n=\pm 2$ bcz both are satisfying
 $54 \neq -2$ & $54 = 2$, so 54 is always +ive

⇒ Sum of infinite GP: It will be only possible only if
 $|r| < 1$ i.e. r is in fractions
 unless sum will be actually infinite.



$$|r| < 1 \implies -1 < r < 1$$

$$S = ar + ar^2 + ar^3 + \dots$$

$$rS = ar^2 + ar^3 + ar^4 + \dots$$

$$rS = ar^2 + ar^3 + ar^4 + \dots$$

$$S - rS = a$$

So

$$S = \frac{a}{1-r}$$

, $|r| < 1$, sum of inf. GP.

Also can be derived from general formulae.

Now, then (Imp Ambiguity)

$n=2$ (given)

Step ①

$$n^2 = 4$$

[$n > 0$ need to put]

Step ②

$$n = \pm \sqrt{4}$$

$$n = \pm 2$$

So we were given $n=2$ & suddenly we are saying n is ± 2 . It is not possible. Something is wrong.
 Wrong is step ①, we cannot sq. directly, we can square by putting condition $n > 0$ on step - ①,
 So now $n = -2$ is N.P. and we have only $n=2$.

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Ambiguity - 2

$$\sqrt{-1} = i$$

$$i^2 = (\sqrt{-1})^2$$

$$i^2 = -1$$

Method - 1

$$i^2 = i * i$$

$$= \sqrt{-1} * \sqrt{-1}$$

$$= \sqrt{-1 * -1}$$

$$= \sqrt{1}$$

$$i^2 = 1$$

Method - 2

Then one of the method here is wrong obviously. Method 2 is wrong bcz $\sqrt{a} * \sqrt{b} \neq \sqrt{ab}$ if a & b both are negative. It is only true i.e. $\sqrt{a} * \sqrt{b} = \sqrt{ab}$ when atleast one of a, b is +ive.

Ques: A football is dropped from 600 cm each time, it rebound it rises $\frac{2}{3}$ rd of the height it has fallen through. Find the total distance travelled by the ball before it comes to rest.

$$\Rightarrow S_n = 600 + \overset{\substack{\text{Coming} \\ \text{back}}}{600 * \frac{2}{3}} + 600 * \left(\frac{2}{3}\right)^2 + \dots$$

Now it is an G.P. with $|r| < 1$

So

$$S_n = \frac{a}{1-r} \Rightarrow \frac{600}{1-\frac{2}{3}} \Rightarrow \frac{600}{\frac{1}{3}} \Rightarrow 1800 \text{ cm}$$

$$S_n = 600 + 2 \left[\frac{2}{3} * 600 + \left(\frac{2}{3}\right)^2 * 600 + \dots \right]$$

$$= 600 + 2 \left[\frac{\frac{2}{3} * 600}{1 - \frac{2}{3}} \right] \quad \left(\begin{array}{l} \text{G.P.} \\ a = \frac{2}{3} * 600 \text{ bcz } \frac{2}{3} \end{array} \right)$$

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$$\Rightarrow 600 + 2 \left[\frac{400}{\frac{1}{5}} \right]$$

$$\Rightarrow 600 + 2(1200)$$

$$\Rightarrow 600 + 2400 \Rightarrow \underline{\underline{3000 \text{ cm.}}}$$

Gate 2023 (Indirectly)

Ques: Find summation of following G.P:

$$n, \frac{n}{2}, \frac{n}{4}$$

$$\Rightarrow \frac{n}{1 - \frac{1}{2}} \Rightarrow \underline{\underline{2n}}$$

(∞ G.P with $a=n, r=\frac{1}{2}$)

★ Arithmetic - Geometric Progression:

1, 2, 3, ..., n is A.P.

$x, x^2, x^3, \dots, x^n$ is G.P.

$1x, 2x^2, 3x^3, \dots, nx^n$ is AGP

(one term in AP & one in GP)

Ques: $1 + 2x + 3x^2 + 4x^3 + \dots$, $|x| < 1$

$$S = 1 + 2x + 3x^2 + 4x^3 + \dots$$

$$Sx = x + 2x^2 + 3x^3 + \dots$$

(Multiply with x of GP)

$$S - Sx = 1 + x + x^2 + \dots$$

$$S(1-x) = 1 + x + x^2 + \dots$$

$$S = \frac{1 + x + x^2 + \dots}{(1-x)}$$

(G.P. ∞) $\infty |x| < 1$

$$S = \frac{\frac{1}{1-x}}{(1-x)}$$

$$S = \frac{1}{(1-x)^2}$$

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Ques: $\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots$

$\Rightarrow$ Here 1, 2, 3, 4 in A.P.

$\frac{1}{2}, \frac{1}{4}, \frac{1}{8}$ are in G.P.

$S = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots$ (let $x = \frac{1}{2}$)

$S = 1x + 2x^2 + 3x^3 + 4x^4 + \dots$

$xS = x^2 + 2x^3 + 3x^4 + \dots$

$S - xS = x + x^2 + x^3 + x^4 + \dots$

$S(1-x) = \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \dots$

$\frac{1}{2}S = \frac{\frac{1}{2}}{1 - \frac{1}{2}}$

$\frac{1}{2}S = \frac{\frac{1}{2}}{\frac{1}{2}}$

$\frac{1}{2}S = 1$

$S = 2$

$S = 2$

Another View

$\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots$

$= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$

$+ \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$

$+ \frac{1}{8} + \frac{1}{16} + \dots$

$+ \frac{1}{16} + \dots$

$\Rightarrow \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$

$= \frac{1}{2} \quad (\text{Using } \infty \text{ GP})$

$= \frac{1}{2}$

$= \frac{1}{4}$

$= \frac{1}{8}$

$-|x| > 0 \Rightarrow |x| < 1$? No.

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How

$$S = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$$

= (Again using ∞ GP = 2)

Ques: $2 + 5x + 8x^2 + 11x^3 + \dots \infty, |x| < 1$

$\Rightarrow S = 2 + 5x + 8x^2 + 11x^3 + \dots \infty$

$xS = 2x + 5x^2 + 8x^3 + \dots$

(multiply by x)

$$\begin{aligned} S - xS &= 2 + 3x + 3x^2 + 3x^3 + \dots \\ &= 2 + 3(x + x^2 + x^3 + \dots) \end{aligned}$$

$$S - xS = 2 + 3\left(\frac{x}{1-x}\right)$$

$$S = \frac{2}{1-x} + \frac{3x}{(1-x)^2}$$

Ques: $1 - 3x + 5x^2 - 7x^3 + \dots \infty, |x| < 1$

$\Rightarrow S = 1 - 3x + 5x^2 - 7x^3 + \dots \infty$

$-xS = -x + 3x^2 - 5x^3 + \dots$

($x = -x$)

$$\begin{aligned} S + xS &= 1 - 2x + 2x^2 - 2x^3 + \dots \\ &= 1 - 2(x - x^2 + x^3 - \dots) \end{aligned}$$

$$S + xS = 1 - 2\left(\frac{x}{1+x}\right)$$

($a = 1, r = -x$)

$$S = \frac{1}{1+x} - \frac{2x}{(1+x)^2}$$

Lecture IV, V

Modular Arithmetic:

→ Divisibility: If a & b are Integers, a divides b if there is an Integer c such that $ac = b$.

Notation $a|b$ means a divides b .

Ex: $3|6$ as $3 \cdot 2 = 6$

Which of the following are true?

$5|1 \rightarrow \text{false}$

$25|5 \rightarrow \text{false}$

$5|0 \rightarrow \text{True}$

$3|2 \rightarrow \text{false}$

$1|5 \rightarrow \text{True}$

$5|25 \rightarrow \text{True}$

$8|5 \rightarrow \text{false}$

$2|4 \rightarrow \text{True}$

→ The division Algorithm: Let ' a ' be an Integer and ' d ' a positive Integer. Then, there are unique Integers ' q ' & ' r ', with $0 \leq r < d$, such that $a = dq + r$.

So $r = a \bmod d$.

↳ Remainder.

Ex. $11 = 3 \cdot 3 + 2$

$-11 = -4 \cdot 3 + 1$

(Not $-11 = -3 \cdot 3 - 2$ as $0 \leq r < d$)

↳ However correct

but for us do not consider

a is divided by m.

→ $a \bmod m$ represents the remainder when a is divided by m .→ If two numbers a & b leave the same remainder when divided by m then we can say " a is congruent to b , modulo m " & write:

$$a \equiv b \pmod{m}$$

Ex: $5 \equiv 7 \pmod{2}$ ✓

$10 \equiv 15 \pmod{5}$ ✓

$5 \equiv 15 \pmod{3}$ is false.

$8 \equiv 18 \pmod{5}$ ✓

$5 \equiv -2 \pmod{3}$ is false.

→ If we add m in a then rem. will not change.

Ex: $7 \bmod 5 = 2$ always

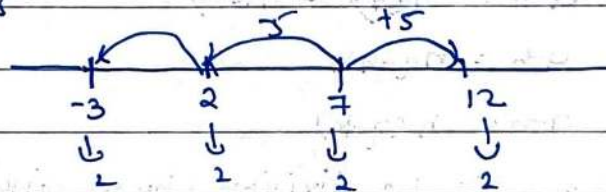
$(7+5) \bmod 5 = 2$

$(7+5x) \bmod 5 = 2$

$(7-5) \bmod 5 = 2$

$(7-5x) \bmod 5 = 2$

mod 5



a, $a \% m = r$ then, $(\% \rightarrow \bmod)$

$$(a + xm) \% m = a \% m + xm \% m$$

$$= r + 0$$

$$= r$$

So, if $a \equiv b \pmod{m}$ is given then,

$$a + xm \equiv b \pmod{m} \quad \&$$

$$\boxed{a + xm \equiv b + ym \pmod{m}}$$
, both are True, x, y are Integer

 $a \equiv a \pmod{m}$ for any $a, m \in \mathbb{Z}$ is true & $a \equiv b \pmod{1}$ for any $a, b \in \mathbb{Z}$ is true as only possible rem is 0.

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→ $x \equiv 0 \pmod{2}$ is true when x is multiple of 2 (i.e. x is ^{even}).

→ if $x \pmod{m}$ then possible remainders are $0, 1, 2, \dots, m-1$.

→ $x \equiv 16 \pmod{10}$ means x should have remainder 16 of 10 i.e. 6 . So then $x = 6 + 10k$ where $k \in \mathbb{Z}$.

→ So $a \equiv b \pmod{m}$ then $a = (b \div m) \cdot m + mk \quad \forall k \in \mathbb{Z}$

→ If $a = n \cdot q + r$ &
 $b = n \cdot k + r$ then,
 $a \equiv b \pmod{n}$ (or vice-versa)

i.e. If $a \equiv b \pmod{n}$

then $a = n \cdot q + r$ & $b = n \cdot k + r$

$$a - b = n \cdot q - n \cdot k$$

From this $a - b = nq - nk$

$$a - b = n(q - k)$$

$$a - b = n(q - k)$$

So, $a - b$ will be divisible by n with quotient $q - k$.

i.e. if a & b leave same remainder then $a - b$ or $b - a$ will not leave any remainder.

$$\rightarrow a \equiv b \pmod{m}$$

$$\text{then, } m \mid a - b$$

$$a \equiv b \pmod{m} \leftrightarrow m \mid a - b \leftrightarrow a \pmod{m} = b \pmod{m}$$

$7 \equiv 4 \pmod{3}$ is $\checkmark$

$7 \equiv 4 \pmod{3}$ is $\times$ as it is $7 \equiv 1$ which is $\times$

$\Rightarrow$ If $a \equiv b \pmod{m}$ means a & b leaves same remainder on dividing with m then $a \pmod{m} \equiv b \pmod{m}$ means $a \pmod{m}$ & b leaves same remainder on dividing with m and which is obviously equivalent to $a \equiv b \pmod{m}$.

$$\Rightarrow a \pmod{m} = (a \pmod{m}) \pmod{m}$$

Ques: T/F?

a) $a \equiv a \pmod{n} \rightarrow \text{True}$

b) $a \equiv -a \pmod{n} \rightarrow \text{False}$

c) If $a \equiv b \pmod{n}$ then $b \equiv a \pmod{n} \rightarrow \text{True}$

d) If $a \equiv b \pmod{n}$ & $b \equiv c \pmod{n}$, then $a \equiv c \pmod{n} \rightarrow \text{True}$

$\Rightarrow$ If $a \equiv b \pmod{m}$, & $c \equiv d \pmod{m}$, then

$a+c \equiv b+d \pmod{m}$ & (Th-1)

$a*c \equiv b*d \pmod{m}$ (Th-2) (Div will not work.)

Also

$a+k \equiv b+k \pmod{m}$ bcz $k \equiv k \pmod{m}$ (k is Integer)

$a+m \equiv b \pmod{m}$ bcz $m \equiv 0 \pmod{m}$

$\hookrightarrow$ can be km

Adding any multiple of m at any of the side will not basically change the remainder so allowed to do so.

Ques: If $a \equiv b \pmod{m}$ then.

a) $a+m \equiv b \pmod{m} \rightarrow \text{True}$

b) $a+3m \equiv b+2m \pmod{m} \rightarrow \text{True}$

c) $a+k \equiv b+m \pmod{m} \rightarrow \text{False}$

d) $a+k+3m \equiv b+k \pmod{m} \rightarrow \text{True}$

e) $a+2k \equiv b+k \pmod{m} \rightarrow \text{False}$

Proof of Theorem-1,2 (Not Req.)

$$\begin{aligned} c \equiv d \pmod{m} &\Rightarrow a+c \equiv b+d \pmod{m}, \text{ & } \\ a \equiv b \pmod{m} &\quad ac \equiv bd \pmod{m} \end{aligned}$$

$\Rightarrow$ We can say

$$m \mid c-d \text{ & } m \mid a-b$$

So also m will divide both's sum

$$m \mid c-d+a-b$$

$$m \mid a+c-(d+b) \Rightarrow a+c \equiv b+d \pmod{m} \quad \checkmark$$

Also m will divide both's product

$$m \mid (c-d)(a-b)$$

$$m \mid ac-bd$$

$$\Rightarrow ac \pmod{m} \Rightarrow (a \pmod{m} \cdot c \pmod{m}) \pmod{m}$$

Gen. convention of congruence

$$\boxed{x \equiv a \pmod{m} \Rightarrow \text{generally } 0 \leq a < m}$$

Div. in th-2 do not work

Ex: $\frac{16}{8} \equiv \frac{6}{3} \pmod{10}$ False

$$2 \equiv 2 \pmod{10} \quad \underline{\text{False}}$$

[Generally Div. is allowed when GCD of 10 & no. div with is 1 (Not Req.)]



Optional
Que:

Solve $3x+4 \equiv 2x+8 \pmod{9}$

 $\Rightarrow$ Subtract $2x$ from both sides

$$x+4 \equiv 8 \pmod{9}$$

Subtract 4 from both sides

$$x \equiv 4 \pmod{9}$$

So $x = 4 + 9k$

Que: Reduce $100^5 \pmod{7}$ to an element in the standard residue system $\Rightarrow$ We need to find the remainder of $100^5 \pmod{7}$ will be greater than 7 so do again mod:

eq. to $(100 \pmod{7})^5 \pmod{7}$

$$\Rightarrow (2)^5 \pmod{7}$$

$$\Rightarrow 32 \pmod{7}$$

$$\Rightarrow \underline{\underline{4}}$$

Que: Simplify $994 \cdot 996 \cdot 997 \cdot 998 \pmod{1000}$ to a no. in range $\{0, 1, \dots, 999\}$ $\Rightarrow$ Can also do it via multiplication by

$$994 \pmod{1000} \cdot 996 \pmod{1000} \cdot 997 \pmod{1000} \cdot 998 \pmod{1000}$$

$$\Rightarrow -6 \pmod{1000} \cdot -4 \pmod{1000} \cdot -3 \pmod{1000} \cdot -2 \pmod{1000}$$

$$\Rightarrow -6 \cdot -4 \cdot -3 \cdot -2 \pmod{1000}$$

$$\Rightarrow \underline{\underline{144}}$$

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Ques: Simplify $17^{753} \pmod{9}$?

$$\begin{aligned}
 \Rightarrow & (17 \pmod{9})^{753} \pmod{9} \\
 & (8 \pmod{9})^{753} \pmod{9} \\
 & (-1 \pmod{9})^{753} \pmod{9} \quad (8 \equiv -1 \pmod{9}) \\
 \Rightarrow & -1 \pmod{9} \\
 \therefore & -1 \pmod{9} = \underline{\underline{8}}
 \end{aligned}$$

GATE CSE 2016Ques: $13^{99} \pmod{17}$

$$\Rightarrow 13^{99} \pmod{17}$$

$$\Rightarrow 13 \cdot (13)^{98} \pmod{17}$$

$$\Rightarrow 13 (13^4)^{24} \pmod{17}$$

$$\Rightarrow 13 (169)^{24} \pmod{17}$$

$$\Rightarrow 13 (1)^{24} \pmod{17}$$

$$\Rightarrow -13 \pmod{17}$$

$$\Rightarrow \underline{\underline{4}}$$

$$\text{as } 169 \equiv -1 \pmod{17}$$

169 & -1 are same in world of mod 17.

GATE CSE 2019Ques:

$$3^{51} \pmod{5}$$

$$\Rightarrow 3 \cdot (3^2)^{25} \pmod{5}$$

$$\Rightarrow 3 (9)^{25} \pmod{5}$$

$$\Rightarrow 3 (-1)^{25} \pmod{5}$$

$$\Rightarrow -3 \pmod{5}$$

$$\Rightarrow \underline{\underline{2}}$$

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Ques: $17 \times 18 \pmod{19}$

$\Rightarrow 17 \pmod{19} \times 18 \pmod{19}$

$\Rightarrow -2 \times -1$

$\Rightarrow +2 \pmod{19}$

$= \underline{\underline{2}}$

Ques. $18^{409612498312456} \pmod{19}$

$\Rightarrow (-1)^{\text{—————}6} \pmod{19}$

$\Rightarrow \underline{\underline{1}}$

Ques. Find the last digit of 7^{100}

$\Rightarrow$ i.e. $7^{100} \pmod{10}$

$\Rightarrow (-3)^{100} \pmod{10}$

$\Rightarrow 9^{50} \pmod{10}$

$\Rightarrow (-1)^{50} \pmod{10}$

$\Rightarrow \underline{\underline{1}}$

Ques. Last digit of 2017^{2017}

i.e. $(2017)^{2017} \pmod{10}$

$(7)^{2017} \pmod{10}$

$\Rightarrow 7 (-3)^{2016} \pmod{10}$

$\Rightarrow 7 (9)^{1008} \pmod{10}$

$\Rightarrow 7 (1)^{1008} \pmod{10}$

$\Rightarrow \underline{\underline{7}}$

NOTE/Thorem: A number is divisible by 9 if and only if sum of its digits is divisible by 9.

$\Rightarrow$ I did it in my child hood ($2+7+7+2=9$ is divisible)

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Proof: To prove that $abcd \equiv a+b+c+d \pmod{9}$

Proof: $abcd = a000 + b00 + c0 + d$
 $(a \times 10^3 + b \times 10^2 + c \times 10 + d) \pmod{9}$
 $\Rightarrow (a \times 10^3) \pmod{9} + (b \times 10^2) \pmod{9} + (c \times 10) \pmod{9} + d \pmod{9}$
 $= a \pmod{9} + b \pmod{9} + c \pmod{9} + d \pmod{9}$
 $= a+b+c+d \pmod{9}$

Hence proved
 (Recursive Hai)

Something little same is for 11 too i.e.

$abcd \pmod{11} = -a+b-c+d \pmod{11}$ ✓
 ↪ Start from this cd , then $-c$ & so on.

Lecture - 6, 7, 8, 9

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★ Proof Techniques:⇒ Disclaimer:

GATE is objective exam, then req. of proof techniques:

- To improve analytical skills
- To improve problem solving skills
- To feel good.
- For Utech/MS interviews

⇒ Conjecture → Statements which are not yet proven nor disproven
↳ Open Statement

Ex: Goldbach's Conjecture i.e. ~~Sum~~ any integer $\geq$ even natural no. > 2 is sum of two prime numbers.
is never proven nor disproven

Ex: like $4 = 2+2$, $6 = 3+3$, $8 = 3+5$, $10 = 5+5$, $12 = 5+7$, etc.

Whole world is trying to prove conjectures but it is not easy to prove them by any technique.

(If, you do so, you are a PhD).

So coming up with proof is hard and not required, but understanding proofs is our requirement. (Don't be demotivated)

⇒ Terminology:⇒ Definition: Something Given (No proof)

like definition of even no. is Integer multiple of 2.

⇒ Theorem: Something to be proven (or already proven).
Proven statement by Mathematicians already.⇒ Axiom: Statements that are always true (No proof needed)

Ex: Given two distinct points, there is exactly one line containing them.

$n=2k \rightarrow n, \text{ even Integer}, k \rightarrow \text{Integer}$

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$n=2k+1 \rightarrow n \text{ is odd integer}, k \text{ any Integer}$

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(Seed)

(Tree)

With small axioms we create great concepts (Seen in prob.)

One Axiom of Real no. is that for all real number x, y , $x+y$ is a real number.

Three axioms of probability are there on which whole prob. is based.

"Axioms are true" is an Axiom.

→ Basic Proof Techniques are there which can be taught, ^(we will learn) but they cannot prove everything in this world. Some problems have their own techniques.

Proof Tech: I $\Rightarrow$ Direct Proof

Template

To prove: If 'p' happens then 'q' happens

Proof: Assume 'p' happened

↓
Apply the facts you know
(logical Deductions)

Derive 'q'.

Ex: W.K.T. even number is an integer which is multiple of 2. Prove that every integer which is multiple of some even no. is even no.

(i.e. Even Int * ^{Any} Int. = Even Int.)

⇒ To prove: If a is even int then ab is even where b is any integer.

Proof: a is even (given) (p)

i.e. $a = 2k$

$ab = 2k \cdot b$

$ab = 2(kb)$

which means ab is even (q)

∴ Hence Proved.

Ex:2 Show that every integer divisible by 6 is divisible by 3.

⇒ To prove: If $6|a$ then $3|a$

Proof: 6 divides a (given)

as 3 divides 6

then, 3 divides a

∴ (Hence Proved)

OR

6 divides a (given)

$a = 6k$

$a = 3 \cdot 2k$

$a = 3(2k)$

↳ Integer

$a = 3i$

that means 3 divides a

∴ Hence Proved.

$a/b \neq a \setminus b$
 $\hookrightarrow$ divided by b (Integer Result)
 $\hookrightarrow$ divide symbol a divides b. (Boolean Result)
 $b = ak, k \in \mathbb{Z}$

$0/0$ is Indeterminate, but
 $0 \setminus 0$ is True
 $\hookrightarrow$ zero divides zero
 $0/5 \rightarrow \text{false}, 5 \setminus 0 \rightarrow \text{True}$

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Theorem: If a & b are consecutive integers, then the sum $a+b$ is odd.

To prove: $a+b = 2k+1$

Proof: Given $b = a+1$ (consecutive integers)

then $a+b = a+a+1$

$$= 2a+1$$

and $2a+1$ is form of odd

$\therefore$ Hence Proved.

Theorem: Sum of two even integers is even.

To Prove: $2p+2q = 2k$

Proof: Sum: $2p+2q$

$$= 2(p+q)$$

$$= 2k$$

$$(k = p+q)$$

$\therefore$ Hence Proved.

Perfect Square Integer Definition: a is P.S. iff $a = b^2$, b is an Integer

Ex: 4, 1, 9, 25, etc.

Prove: If $5|2a$, $a \in \mathbb{R}$ then $5|a$

Proof: $5|2a \Rightarrow 2a = 5k$

$2a \rightarrow \text{even}, 5 \rightarrow \text{odd}$ then k is even

$$2a = 5(2b)$$

$$(k = 2b)$$

$$a = 5b$$

$$5|a$$

$\therefore$ Hence Proved

will be
taught
in P.L.

Proof Tech. II $\rightarrow$ Proof by Contraposition $\Rightarrow$ This is an Indirect Proof.

Template To prove: If 'p' then 'q'

Proof: q is false. (Assume)

Apply the facts
you know

Therefore, p is false (Derive)

True $\Rightarrow$ If p then q is equivalent to If $\neg q$ then $\neg p$.
(Not eq. to If q then p), (Not eq. to If $\neg p$ then $\neg q$)

Ex: If you are India's PM then you are an Indian.
eq. to: If you are not an Indian, then you are not India's PM.
Xb) If you are Indian, then you are India's PM.
Xc) If you are India's PM, then you are not Indian.

So If p, then q $\equiv$ If not q, then not p.

This is how Proof by Contraposition is working.

(If $\neg q$ then $\neg p$ $\neq$ If p then q is direct proof)

Ex: If n^2 is even, then n is even.

$\Rightarrow$ Let n is odd.

$$n = 2k + 1$$

$$n^2 = 4k^2 + 4k + 1$$

$$n^2 = 2(2k^2 + 2k) + 1$$

$$n^2 = 2m + 1$$

$\therefore$ which is not even

So Proved by Contraposition If n is not even
 n^2 is not even.

$\left\{ \begin{array}{l} \text{Even} * \text{Any} = \text{Even} \\ \text{Odd} * \text{Odd} = \text{Odd} \\ \text{Odd} + \text{odd} = \text{Even} \\ \text{odd} + \text{even} = \text{odd} \end{array} \right\} \rightarrow \text{Already Proven}$

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Ex: If $3n+2$ is odd then n is odd

Direct Proof

$$3n+2 = 2k+1 \quad (\text{Given})$$

$$3n+1 = 2k$$

$$3n = 2k-1$$

$$n = \frac{2k-1}{3} \quad ?$$

Stuck. (Can do but lengthy)

Indirect Proof (P.O. Contradiction)

Let n be even not odd (r.p)

$$n = 2k$$

$$\text{then } 3n = 6k$$

$$3n+2 = 6k+2$$

$$3n+2 = 2(k+1) \rightarrow \text{Int}$$

$$3n+2 = 2k \rightarrow \text{Even}$$

So it is not odd (r.p)

Hence Proved.

$\rightarrow$ Negation of $P \wedge Q$ acc. to general intuition is $\text{Not } P \vee \text{Not } Q$.

$\rightarrow$ Negation of $P \vee Q$ is $\text{Not } P \wedge \text{Not } Q$ or None (r.p or r.q)

Ex: Show that if $n=ab$ where a & b are tive Int, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$

$\rightarrow$ we will be stuck by direct proof, let's use Indirect.

$$\text{Let } \neg (a \leq \sqrt{n} \text{ or } b \leq \sqrt{n})$$

$$(\neg(P \vee Q)) = \neg P \wedge \neg Q$$

$$\text{i.e. } a > \sqrt{n} \text{ and } b > \sqrt{n}$$

$$\text{then } a \cdot b > n$$

$$\text{which is not } n=ab$$

$\therefore$ Hence Proved.

Zero $\rightarrow$ neither +ve nor -ve

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★ Rational No.: $\frac{Int}{Int} \rightarrow \neq 0$ Ex: 2, $\frac{3}{2}$, 4.5, $\frac{0}{1}$ (0), 0.0000001

10.6666666... ($\frac{2}{3}$) ; $\frac{22}{7}$

0.12121212... is rational

(Trick) $\hookrightarrow$ let $x = 0.121212...$ pattern is present
Not Req.

$$100x = 12.121212$$

$$100x - x = 12$$

$$x = 12/99$$

★ Irrational No.: No. which are not rational

Ex: 12.1100123459828757596666...

$\hookrightarrow$ Random No. (No pattern)

$\sqrt{2}, \sqrt{3}, e, \sqrt{\text{prime No}}, \pi$

$\pi \neq \frac{22}{7}$ (it is approx)

$\rightarrow$ Sum of two Rational No. is Rational No.

$$\frac{a}{b} + \frac{c}{d} = \frac{Ptda}{qb} \rightarrow \neq 0 \text{ (obviously)}$$

$\rightarrow$ If x is rational then $\sqrt{x}$ can be irrational.

Ex: 2 is Rational but $\sqrt{2}$ is not.

$\rightarrow$ If x is irrational, then $\sqrt{x}$ will be irrational.

Proof by Contraposition:

Let $\sqrt{x}$ be rational

$$\text{then } \sqrt{x} = \frac{p}{q}$$

$$x = \frac{p^2}{q^2}$$

x is Rational (not irrational)

So, if $\sqrt{x}$ is rational $\therefore$ Hence Proved

My shortcut

Proof Tech III: Proof by Contradiction (C.D.) (Most Interesting)

As Sherlock Holmes said, "When you have eliminated the impossible, whatever remains, must be the truth".

i.e. If you can prove that "P is false" is impossible, then you conclude P must be true.

Template:

To prove : P

Proof by C.D.: Prove 'P is false' is impossible

Assume P is false

Use facts

Prove some contradiction & say P is false is impossible

Lowest form of Rational No.:If p/q is rational no, then only common factor of p, q is1. then it is lowest form ($\text{GCD}(p, q) = 1$)Ex- $\frac{2}{3}$ is lowest form, $\frac{12}{18}$ is not lowest form $\frac{6}{1}$ is lowest form, $\frac{6}{8}$ is not lowest formQues: Prove by C.D. " $\sqrt{2}$ is irrational" $\Rightarrow$ Let $\sqrt{2}$ is rationalThen $\sqrt{2} = \frac{a}{b}$, $b \neq 0$ & $\text{GCD}(a, b) = 1$ (lowest form)

$$a^2 = 2b^2$$

 $(a^2 \text{ is even} \Rightarrow a \text{ is even} \Rightarrow a = 2k)$

$$(2k)^2 = 2b^2$$

$$4k^2 = 2b^2$$

$$b^2 = 2k^2$$

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This make b^2 even, then b is even.

If a & b are even then one of the $\text{GCD}(a, b)$ is 2.
which is not possible.

↳ It is contradiction bcz a, b are in lowest form

So $\sqrt{2}$ is rational is not possible.

So $\sqrt{2}$ is irrational.

∴ Hence **Proved**.

Ques: Prove that there is no largest Integer.

⇒ First, ∞ is not no., It is just a concept.

So, can't say ∞ is largest no. bcz ∞ is not even a no.

Proof: Assume a is largest Integer

then $a+1$?

↳ Contradiction

So Not possible So No largest Integer.

Ques: Show that at least four of any 22 days must fall on the same day of the week.

⇒ Let no day falls four times in 22 days.

then all 7 days fall ^{at most} three times ~~at least once~~

then, total days = 7×3 (Max)

$= 21 \neq 22$

↳ contradiction

∴ Hence **Proved**.

Proof tech. IV: Disproof by Counterexample:

Till now we tried to prove P . Here we will disprove it by giving a counter example.

Ex: Let P : Every actor is Indian
then & if Keanu Reeves ~~prove~~ disproves P .

Ex: Prove or disprove the conjecture,
"for every positive integer n , $n! \leq n^2$ ".

Let $n=4$, then $n! = 4 \times 3 \times 2 = 24$

$$n^2 = 16$$

$$\text{So } n! > n^2$$

So $\therefore$ Hence disproved by counter example.

Ex: If a divides bc , then $a|b$ or $a|c$, $a, b, c \in \mathbb{Z}$

$\Rightarrow$ False Let $a=6$, $b=2$, $c=3$
then $a|bc$ but a ~~don't~~ divide b or c .

$\therefore$ Disproved.

NOTE Euclid's Lemma: If prime no. P divides bc , then
either $P|b$ or $P|c$.

(Proof is complicated, So skip).

$\rightarrow$ Proof by Induction is not imp. for GATE perspective
Also, it can only prove if we already know the
proof result.

(M.E) (Not much req. to us)

Proof Tech. V: Proof using Mathematical Induction: (Easy / fir process)

Intuition behind it is by relating it with Domino effect. Why all dominoes fall?

→ We push first domino & then every k^{th} domino pushes $k+1^{\text{st}}$ domino, for every $k \geq 1$



↳ n can be large no.,

but not ∞ , as ∞ is not no., just a concept.

So, the principle of M.P. states that if for some property $P(n)$, we have that

It starts $\rightarrow P(0)$ is true, and.

and it keeps going

for any natural no n , $P(n) \rightarrow P(n+1)$

Then,

for any natural no n , $P(n)$ is true.

So

$P(1) \rightarrow P(2) \rightarrow P(3) \dots$

↓
manually
prove

So, for every natural no.,

$P(n)$ is true

Ex: 1 Prove $1+2+3+\dots+n = \frac{n(n+1)}{2}$

→ To prove: $P(n)$ for any $n \in \mathbb{N}$

$$\begin{array}{ccc} P(1) \text{ is } 1 & \frac{1(1+1)}{2} = \frac{2}{2} = 1 & \\ \text{LHS} & \text{RHS} & \checkmark \end{array}$$

Here MI is only for Natural Nos (N)

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Now let's assume $P(k)$ is true:

$$\text{i.e. } 1+2+3+\dots+k = \frac{k(k+1)}{2}$$

then

$$P(k+1) = 1+2+3+\dots+k+k+1$$

$$= \frac{k(k+1)}{2} + (k+1)$$

$$= \frac{k(k+1)+2k+2}{2} \Rightarrow \frac{k^2+k+2k+2}{2}$$

$$\Rightarrow \frac{k^2+3k+2}{2}$$

$$\Rightarrow \frac{k^2+2k+k+2}{2} = \frac{k(k+1)+1(k+1)}{2}$$

$$P(k+1) \Rightarrow \frac{(k+1)(k+2)}{2}$$

= R.H.S for $P(k+1)$

So we can go from $P(k)$ to $P(k+1)$

& we have also started by $P(1)$

So, R.H.S is true for any Natural No.

So by principle of MI. Hence Proved.

Ex:2

Use MI to show that for all non-neg Integers n
 $1+2+2^2+\dots+2^n = 2^{n+1}-1$

$\Rightarrow$ To prove $\forall P(n)$ for $n \in \mathbb{N}$

$$P(0) \Rightarrow 1 = 2^1 - 1$$

$$1 = 1$$

(Here base case is $P(0)$ not $P(1)$.)

Let's assume $P(k)$ is true

i.e. $1+2^1+2^2+\dots+2^k = 2^{k+1}-1$

Then

$$\begin{aligned} P(k+1) &= 1+2^1+2^2+\dots+2^k+2^{k+1} = 2^{k+1}-1+2^{k+1} \\ &= 2^{k+1}-1+2^{k+1} \\ &= 2 \cdot (2^{k+1}) - 1 \\ &= 2^{k+2} - 1 \\ &= R.H.S. \end{aligned}$$

So by principle of M.I. $P(n)$ is true $\forall n \in \mathbb{Z}^+$ as we have base case and we can go from $P(k)$ to $P(k+1)$.

Hence Proved.

Ques: Prove the following proposition:

Suppose $a, b, c \in \mathbb{Z}$. If $a^2+b^2=c^2$, then a or b is even.

$\Rightarrow$ Proof by C.D (not C.P)

Let a & b are both odd.

then a^2 & b^2 both are odd $\because$ odd² $\rightarrow$ odd.

and then a^2+b^2 is even $\because$ odd+odd $\rightarrow$ even

So c^2 is even then

c is even $\because$ even² $\rightarrow$ even

So let $c = 2k$, $a = 2m+1$ & $b = 2n+1$

$$(2m+1)^2 + (2n+1)^2 = 4k^2$$

$$4m^2 + 4m + 1 + 4n^2 + 4n + 1 = 4k^2$$

$$2m^2 + 2m + 2n^2 + 2n + 1 = 2k^2$$

$$\underbrace{2m^2 + 2m + 2n^2 + 2n}_{\text{Even}} + \underbrace{1}_{\text{Odd}} = \underbrace{2k^2}_{\text{Even}}$$

So, a & b both cannot be odd

So there is contradiction if a is not True then there comes contradiction in P . So P is correct. $\therefore$ Hence Proved.

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An Interesting consequence of this result is that in any right angle triangle in which all sides have integer length, then at least one of the two shorter sides must be of even length.

(because $\text{longer}^2 = \text{shorter}_1^2 + \text{shorter}_2^2$)

And many more imp applications of this statements like sum of two odd perfect squares is never a perfect square.

→ Prime Numbers: These positive integers which are divisible by exactly two positive integers (1 & itself).

Ex: 2, 3, 5, 7, 11, etc. (1 is not prime bcz it is div- by one integer only).

Ques: Consider the following statement, for every prime number p , either $p=2$ or p is odd. Now try to prove it.

→ Proof by Contraposition

⇒ Let $p \neq 2$ and p is not odd. ~~Atx~~ ~~Contr~~

then $p = 2k$, $k \neq 1$

then $\frac{p}{2} = k$ & $k \neq 1$

So, p got new divisors i.e. $\frac{p}{2}$, which is neither giving p nor 1.

So p is not prime no.

So by Contraposition, either $p=2$ or p is odd should be true.

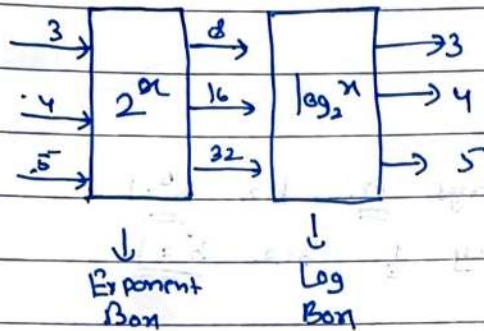
∴ Hence Proved.

Lecture-10

★ Logarithms

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These are another way of thinking about a^x .

So we can call log as inverse of exponent function

In a^x and $\log_a y$, a is exponent base & log base respectively.

So

$$\log_b n = y \iff n = b^y$$

Solve using just definition:

a) $\log_{10} 10000$: Think what power of 10 brings 10,000 i.e.

$$\text{So} = \underline{4}$$

$$\text{b) } \log_5 \left(\frac{1}{125} \right) = \underline{-3}$$

$$\text{let } \log_5 (1/125) = y$$

$$\text{then } 5^y = \frac{1}{125}$$

$$= 5^y = 5^{-3}$$

$$\text{So, } y = -3$$

$$\text{c) } \log_{1/2} \left(\frac{1}{8} \right)$$

$$\text{let } y = \log_{1/2} \left(\frac{1}{8} \right)$$

$$\left(\frac{1}{2} \right)^y = \frac{1}{8} \Rightarrow y = \underline{3}$$

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d) $\log_4 1$

$\Rightarrow \text{let } y = \log_4 1$

then $4^y = 1$

$4^0 = 1$

$y = 0$

which means $\log_b 1$ is always 0, bcz $b^0 = 1$ and also $\log_b b$ is always 1 bcz $b^1 = b$.

So,

$\log_b b = 1 \text{ \& } \log_b 1 = 0$

→ obs-①

→ obs-②

e) $\log_2 0$

$\Rightarrow \text{let } y = \log_2 0$

then $2^y = 0$

but it is not possible. no finite value of y ,So No soln. (Undefined)

f) $\log_3 -1$

$\Rightarrow \text{let } y = \log_3 -1$

so $3^y = -1$

Since 3 is a +ve no. any power of it cannot lead us to negative no. So no y exists here also.So No soln. (Undefined)So zero & negative can't be put in $\log$. It is function of positive numbers.

$$b^x = a \Leftrightarrow \log_b a = x$$

fix it in mind

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For a very large n , $\log n$ can be very small.

Ex: if $n = 10^{1000}$ then $\log_{10} 10^{1000} = 1000$, & $1000 \ll 10^{1000}$

So complexity of $\log(n)$ ~~matter~~ ^{very small} as compared that of n .

★ Formal Definition of log: If x & b are positive real no. such that $b \neq 1$, then $y = \log_b x$ is called the logarithmic function with base b & is equivalent to $b^y = x$.

Various bases of logs are possible like $\log_a$, $\log_{10}$, $\log_{100}$, $\log_{12345}$, and many more. But the three most common log bases are 10 , 2 , and e .

→ $\log_{10}$ → Very Common log

→ $\log_e$ → Natural log = $(\ln)$

→ $\log_2$ → Used in CS.

⇒ Some basic properties of logs.

→ $\log_b a^n = n \log_b a$

— P①

→ $\log_b(mn) = \log_b m + \log_b n$

→ P②

→ $\log_b(m/n) = \log_b m - \log_b n$

— P③

⇒ Proof of P②

let $x = \log_b m$ & $y = \log_b n$

then $m = b^x$ & $n = b^y$

L.H.S = $\log_b(mn) = \log_b b^x \cdot b^y$

= $\log_b b^{x+y} \Rightarrow x+y \log_b b$ (by obs ①)

= $x+y \Rightarrow \log_b m + \log_b n$

= R.H.S

⇒ Proof of P③

L.H.S = $\log_b(m/n) = \log_b \left(\frac{b^x}{b^y} \right)$

= $\log_b b^{x-y} = x-y$

= $\log_b m - \log_b n = R.H.S$

∴ Hence Proved

$$(a^b)^n \neq a^{(b^n)} = a^{b^n}$$

$\hookrightarrow a^{b^n}$ because $a^b \cdot a^b \cdot a^b \dots n(\text{times}) : \underline{a^{b^n}}$

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$\Rightarrow$ Proof of P.O

i.e. $\log_b a^n = n \log_b a$

let $x = \log_b a^n$ (1)

$y = \log_b a$ (2)

then to prove: $x = ny$

Also from (1) & (2)

$$b^x = a^n$$

$$\& b^y = a$$

$$\text{s.o. } b^x = (b^y)^n$$

$$\text{So } x = ny$$

$\therefore$ Hence Proved.

$\therefore$ Proofs are no more required.

Ques:

Solve:

$$\begin{aligned} \text{a) } \log_5 (555) &= \log_5 5^{3/2} = 3/2 \log_5 5 \\ &= 3/2 \\ &= \underline{\underline{1.5}} \end{aligned}$$

$$\begin{aligned} \text{b) } \log_2 (1/8) &= \log_2 2^{-3} \Rightarrow -3 \log_2 2 \\ &= \underline{\underline{-3}} \end{aligned}$$

Some more properties:

$$\rightarrow a = b^{\log_b a} \quad - \text{ P (14) } \quad (\text{Used too much})$$

$\hookrightarrow$ Remember.

$$\rightarrow b^{\log_b a} = a \quad - \text{ P (15) }$$

(Basically $b^{\log_b a} = a$ is due to this cancellation - Way to remember)

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Proof:

$$a = b^{\log_b a}$$

take $\log_b$ both sides

$$\log_b a = \log_b b^{\log_b a}$$

$$\log_b a = \log_b a \cdot \log_b b$$

$$\log_b a = \log_b a$$

$$a = a$$

$$\text{Ex: } \log_{10} 10^{\log_{10} \pi} = \pi$$

Imp:

$$\log_a x = \frac{\log_b x}{\log_b a}$$

- P (VI) (Change of base property)

→ Important property to change log base

Proof:

$$\text{let } \log_a x = y$$

(LHS)

$$x = a^y$$

$$\text{R.H.S: } \frac{\log_b a^y}{\log_b a} = y \frac{\log_b a}{\log_b a} \Rightarrow y = \text{L.H.S}$$

∴ Hence Proved

$$\text{Ex: } \rightarrow \log_b a = \frac{1}{\log_a b}$$

- P (VII)

Proof:

$$\log_b a = \frac{\log_a a}{\log_a b}$$

(Property II - New base is a)

$$= \frac{1}{\log_a b}$$

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Ques: T/F? $\log_{1/\sqrt{b}} \sqrt{x} = \log_b x$

$$\Rightarrow \frac{\log_b \sqrt{x}}{\log_b 1/\sqrt{b}} = \frac{\frac{1}{2} \log_b x}{-\frac{1}{2} \log_b b} = -1 \log_b x = \text{False}$$

Ques: find the value of, $\log_3 4 * \log_4 5 * \log_5 6 * \log_6 7 * \log_7 8 * \log_8 9$

$\Rightarrow$ Change to the common base

$$\frac{\log_{10} 4}{\log_{10} 3} * \frac{\log_{10} 5}{\log_{10} 4} * \frac{\log_{10} 6}{\log_{10} 5} * \frac{\log_{10} 7}{\log_{10} 6} * \frac{\log_{10} 8}{\log_{10} 7} * \frac{\log_{10} 9}{\log_{10} 8}$$

$$\Rightarrow \frac{\log_{10} 9}{\log_{10} 3} = \log_3 9$$

$$\Rightarrow \text{Eq. to change of base property } \log_3 9 \Rightarrow \log_3 3^2 \Rightarrow 2 \log_3 3 \Rightarrow \underline{2}$$

$$\rightarrow a^{\log_b c} = c^{\log_b a} \quad - \text{ P. (VIII)}$$

Proof: Take $\log_b$ both side

$$\log_b a^{\log_b c} = \log_b c^{\log_b a}$$

$$= \log_b c \cdot \log_b a = \log_b a \cdot \log_b c$$

$\therefore$ Hence Proved.

Ques: Solve for the value of x :

$$\log_3 (x-5) + \log_3 (x+3) = 1$$

$$\Rightarrow \log_3 (x-5)(x+3) = 1$$

$$9 = (x-5)(x+3)$$

$$9 = x^2 - 2x - 15$$

$$x^2 - 2x - 24 = 0$$

On solving $x = 6, x = -4$
 $\xrightarrow{\text{Not Defined}}$

Ques: Given $\log_a x = \frac{1}{\alpha}$, $\log_b x = \frac{1}{\beta}$, $\log_c x = \frac{1}{\gamma}$, then $\log_{abc} x$:

a) $\alpha\beta\gamma$

b) $\frac{1}{\alpha\beta\gamma}$

c) $\alpha+\beta+\gamma$

d) $\frac{1}{\alpha+\beta+\gamma}$

$$\Rightarrow \log_{abc} x = \frac{1}{\log_n abc} = \frac{1}{\log_n a + \log_n b + \log_n c} \quad \text{--- (1)}$$

Given $\log_a x = \frac{1}{\alpha} \Rightarrow \log_n a = \alpha$

Similarly $\log_n b = \beta$ & $\log_n c = \gamma$

Using in (1)

$$= \frac{1}{\alpha + \beta + \gamma} \quad \text{(d)}$$

Ques: If $\log_a a$, $a^{1/2}$ & $\log_a x$ are in G.P. then x is:

$\Rightarrow$ As we know property of G.P. that $b^2 = a \cdot c$.

So $(a^{1/2})^2 = \log_a a \cdot \log_a x$

$$a^1 = \frac{\log_a a}{\log_a b} \cdot \frac{\log_a a}{\log_a x} \times \frac{\log_a x}{\log_a b}$$

$$a^1 = \frac{\log_a a}{\log_a b}$$

$$a^1 = \log_b a$$

$$\log_a a^x = \log_a \log_b a$$

(taking $\log_a$ both side)

$$x = \log_a \log_b a \quad \text{(a)}$$

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Ques: If $\frac{\log x}{b-c} = \frac{\log y}{c-a} = \frac{\log z}{a-b}$, then which of the following is correct?

a) $xyz = 1$

b) $x^a y^b z^c = 1$

c) $x^{b+c} y^{c+a} z^{a+b} = 1$

d) All of the above

$\Rightarrow$ Let $\frac{\log x}{b-c} = \frac{\log y}{c-a} = \frac{\log z}{a-b} = k$

$\log x = k(b-c)$
 $x = 10^{k(b-c)}$

Similarly $y = 10^{k(c-a)}$
 $z = 10^{k(a-b)}$

$xyz = 10^{k(b-c)} \cdot 10^{k(c-a)} \cdot 10^{k(a-b)}$
 $= 10^{kb-kc+kc-ka+ka-kb}$

$\Rightarrow = 10^0$

$\Rightarrow xyz = 1$ (A)

$x^a y^b z^c = 10^{ak(b-c)} \cdot 10^{bk(c-a)} \cdot 10^{ck(a-b)}$
 $= 10^{akb-akc+bkc-bka+cka-ckb}$

$= 10^0$

$= 1$ (B)

Similarly (C) will be correct

So (d) All of the above.

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Ques: The no. of meaningful solutions of $\log_2 (x-1) = \log_2 (x-3)$ is

a) 0

b) 1

c) 1

d) 3

$\Rightarrow \log_2 (x-1) = \log_2 (x-3)$

$\frac{1}{2} \log_2 (x-1) = \log_2 (x-3)$

($\log_a C = \frac{1}{b} \log_a C$)

$\log_2 (x-1)^{1/2} = \log_2 (x-3)$

$= (x-1)^{1/2} = (x-3)$

($\log$ is a one-to-one function)

in $\log_a$, $b \neq 1, b > 0$ &
 $a > 0$

$$(x-1) = (x-3)^2$$

$$x-1 = x^2+9-6x$$

$$x^2+9-6x-x+1=0$$

$$x^2-7x+10=0$$

$$x^2-5x-2x+10=0$$

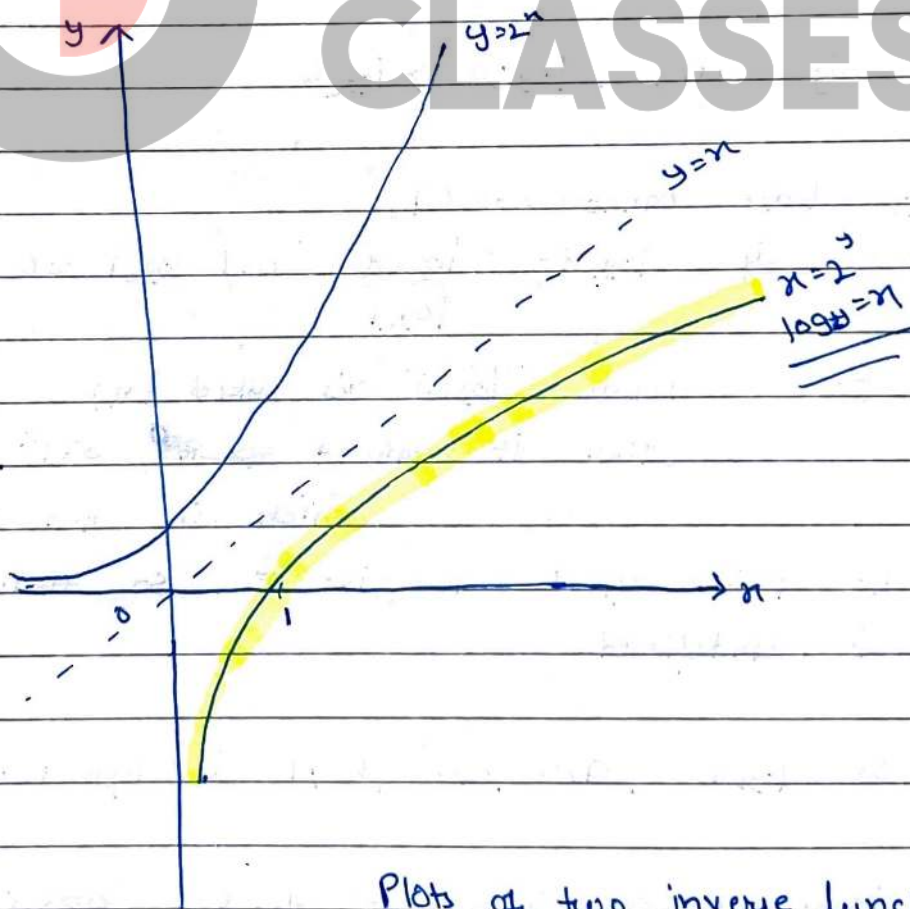
$$x(x-5)-2(x-5)=0$$

$$(x-2)(x-5)=0$$

$x=2, 5$ On putting $x=2$ $\log_2(x-3)$ will be
-ive, which is not defined.

So the meaningful value of x is 5. (i.e. one) (b)

★ Graph for log function:

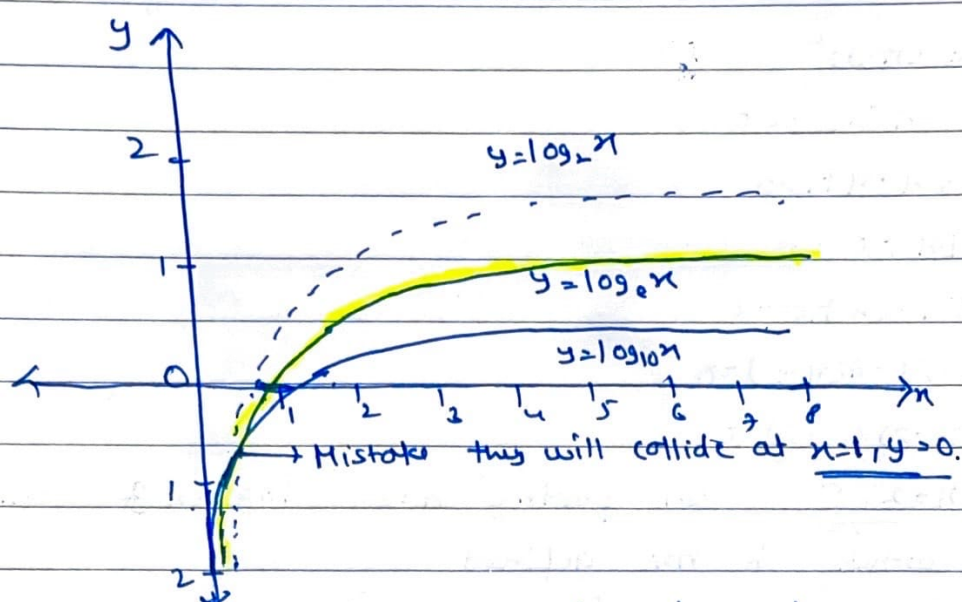


Plots of two inverse function is
mirror image from $y=x$ & $\log_2 y=x$
is inverse of $y=2^x$. Using this
way we can plot graph of log with any base.

logarithm is an anagram of algorithm.

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Graph of $f(x) = \log_b x$ for $b = 2, 10, e$.

So graph of all these are increasing function
(Graph of $\log_{1/2}$ will be decreasing function).

So if $a > b$, $\log_c a > \log_c b$

Ques. Why base cannot be 1?

$\Rightarrow$ bcz. if $\log_c x = \frac{\log_e x}{\log_e c}$ and $\log_e 1 = 0$, So that's why

Or. assume $\log_1 5$ is asked (x)

then it means $\Rightarrow 5 = 1^x$

which is not possible

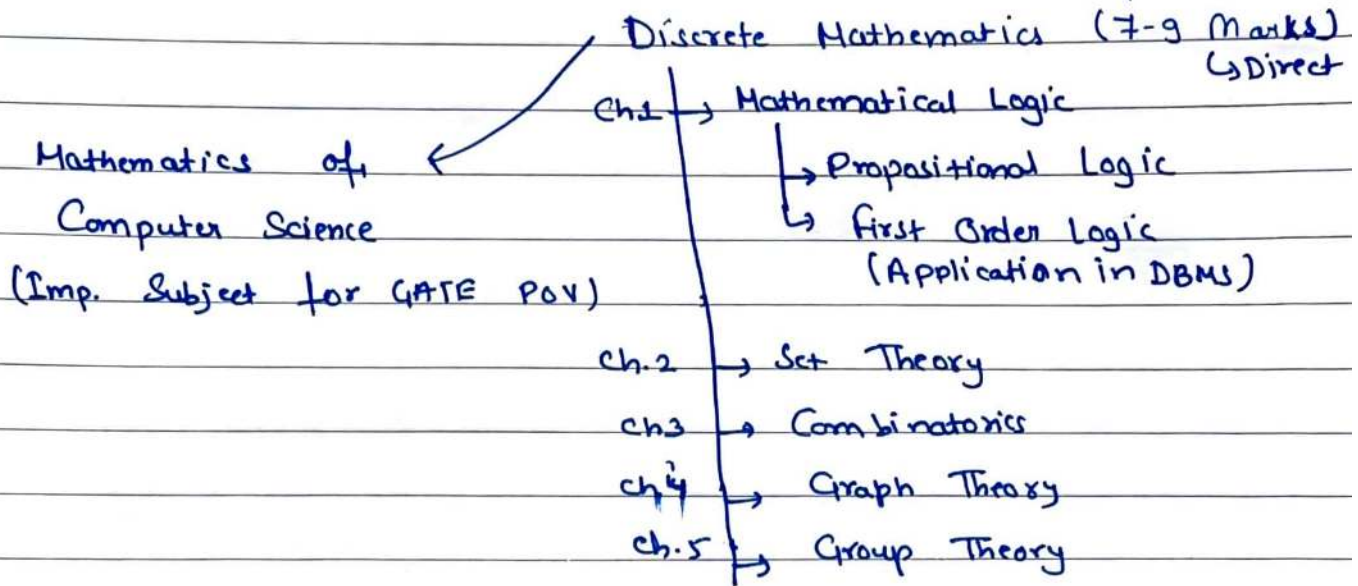
no power of 1 can give 5, So that's why it is undefined.

So $\log_b a$, $a > 0$, $b > 0$, $b \neq 1$ is defined only.

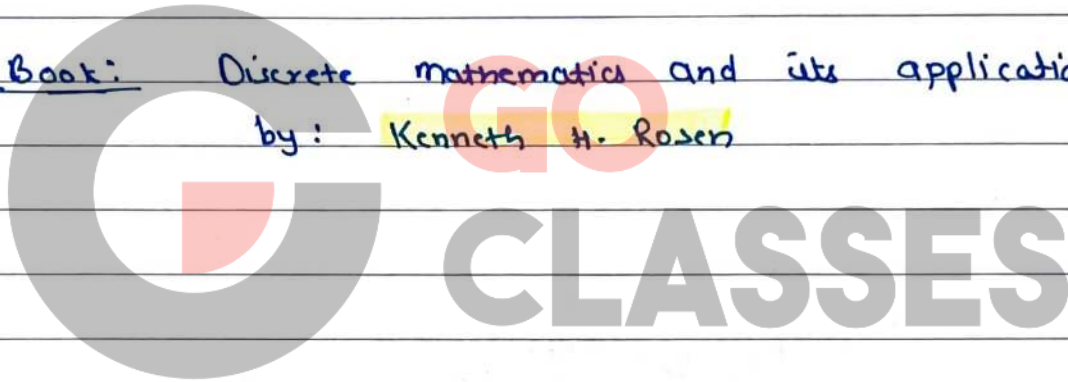
* So logarithm is an imp. part of CS especially log_2

(GATE Marks)

Have weightage Indirectly



Reference Book: Discrete mathematics and its applications
by: Kenneth H. Rosen



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For any questions or issues regarding my handwritten notes, feel free to reach out via email at karan757527@gmail.com or DM me on telegram at @karan757527 (or by clicking [here](#)).

I would love to know if you liked my notes. Your feedback and appreciation are invaluable, so don't hesitate to share your thoughts. Click [here](#) to connect with me on LinkedIn or you can find me as @agrawalkaran.

Wishing you success on your journey ahead!

- Karan Agrawal